

Problem 16.3.15

Evaluate counterclockwise.

$$\oint_C \tan 2\pi z \, dz, \quad C : |z - 0.2| = 0.2$$

Problem 16.3.17

Evaluate counterclockwise.

$$\oint_C \frac{e^z}{\cos z} dz, \quad C : \left| z - \frac{\pi i}{2} \right| = 4.5$$

Problem 16.3.19

Evaluate counterclockwise.

$$\oint_C \frac{\sinh z}{2z - i} dz, \quad C : |z - 2i| = 2$$

Problem 16.3.21

Evaluate counterclockwise.

$$\oint_C \frac{\cos \pi z}{z^5} dz, \quad C : |z| = \frac{1}{2}$$

Problem 16.4.1

Evaluate.

$$\int_0^\pi \frac{2}{k - \cos \theta} d\theta$$

Problem 16.4.7

Evaluate.

$$\int_0^{2\pi} \frac{a}{a - \sin \theta} d\theta$$

Problem 16.4.9

Evaluate.

$$\int_0^{2\pi} \frac{\cos \theta}{13 - 12 \cos 2\theta} d\theta$$

Problem 16.4.11

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx$$

Problem 16.4.15

Evaluate.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6 + 1} dx$$

Problem 16.4.17

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4 + 1} dx$$

Problem 16.4.19

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{x^4 - 1} dx$$

Problem 16.4.21

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)(x^2+4)} dx$$

Problem 16.4.25

Find the Cauchy principal value.

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3-x} dx$$